Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/14/2005
PI Name:	Ju, Yiguang Ph.D.		
Project Title:	STUDIES OF THERMAL RADIATION EFFECTS ON HIGH PRESSURE FLAME DYNAMICS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08540-5263	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	01/12/2004	**End Date:**	09/30/2007
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (12/06)		
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	The effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen mixtures will be measured by a spherical propagating and counterflow flames. The spectral radiation will be detected and analyzed by using an infrared spectrometer and the InSb detector. An optimised radiation model will be developed for spectral radiation prediction by comparing the predicted and measured radiation flux at given spectral wavelength. A drop module will be designed and built. Task Objectives 1. Measure the effect of CO2 addition on flame speed and flammability limit of methane-air and methane-oxygen. 2. Develop a new flame initiation theory.		

Task Description:	3. Measure the the flame spectral radiation by using a spectrometer and InSb detector. 4. Develop an optimised radiation model 5. Design and build a drop module (rig). Task Significance: 1. Provide the first hand data of the effect of CO2 addition on fire control and burning limit. 2. Establish a new model for spectral radiation computation. 3. Provide flame radiation database for microgravity experiment 4. Established a new flame theory for fire suppression.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1. Establish a simplified model for fire initation and extinguishment. 2. Enable modeling of fire spreading in microgravity and normal gravity enviroment. 3. Provide fire detection capability using a CO2 radiation sensor.
Task Progress:	1.1 We have measured the effect of CO2 addition on the flammability limits and flame speeds of CH4-O2-CO2-He mixtures at normal and elevated pressures using the outward propagating flames in a pressure-release type spherical bomb. The measured data were compared with numerical simulations utilizing a detailed emission/absorption statistical narrow band model. It was found that reabsorption can increase burning velocities of CH4-O2-He-CO2 mixtures and extend flammability limits. Furthermore, it is found that radiation reabsorption effect increases with the increase of environmental pressure. It is shown that for optically thick flames with CO2 dilution, reabsorption effects must be considered to correctly predict flame speeds and extinction limits in simulation. The work was presented at 43rd AIAA Aerospace Sciences Meeting and Exhibit on Jan.10, 2005 and at The 20th International Colloquium on the Dynamics of Explosions and Reactive Systems, Aug.1-5, Mcgill, Canada. 1.2 A theory to predict fire initiation from an external energy addition was developed. Emphasis was placed on the effects of the radiation heat loss, ignition power, and mixture Lewis number on the correlation and transition between the initial flame kernel, the self-extinguishing flame, the flame ball, the outward propagating spherical flame, and the planar propagating flame. The resent study extended the previous results by bridging the theories of the non-adiabatic stationary flame balls and traveling flames and allowing rigorous consideration of radiation heat losses. The results showed that the effects of radiation heat loss played an important role on flame regimes and flame transition and resulted in a new isolated self-extinguishing flame. Furthermore, it was found that radiation had significantly reduces the critical ignition radius and resulted in a three different dependences of the minimum ignition power on the Lewis number. Comparisons between the numerical simulations and the analytical results showed a good agreement. The results suggested that prediction of flame initiation without appropriate consideration of radiation is not acceptable. The work is submitted to Combustion Theory and Modeling and will be presented at the 44th AIAA Aerospace Sciences Meeting and Exhibit on Jan.9, 2006. 1.3 We developed a new Statistical Narrow Band-Correlated K (SNBCK) model to simulate the spectral radiation for spherical propagating flames. This algorithm has a better prediction accuracy than the previous SNB model developed by the PI and significantly improved the agreement between the prediction and the experimental results. The radiation model is integrated with a compressible reactive flow code to calculate ignition, flame speed, and flammability limits. This code is one of the most advanced computation codes available in combustion society. The work will be submitted to the next International Symposium on Combustion (2006). 1.4 We have set up an optical measurement system for spectral flame radiation. The system includes a spectrometer (SpectraPro 2500i, 200nm-10um), an infrared Detection assembly (InSb) for 1.5 to 5.0 microns, and a lock in amplifier amplifier (Stanford Research). Flame radiation from counterflow flame has been measured and calibrated. 1.5 We also visited NASA-Glenn Research Center to arrange microgravity experiments in this May.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Ju Y, Xue Y. "Extinction and flame bifurcations of stretched dimethyl ether premixed flames." Proceedings of the Combustion Institute 2005 Jan;30(1):295–301. http://dx.doi.org/10.1016/j.proci.2004.08.258 , Jan-2005
Articles in Peer-reviewed Journals	Qin X, Ju Y. "Measurements of burning velocities of dimethyl ether and air premixed flames at elevated pressures." Proceedings of the Combustion Institute 2005 Jan;30(1): 233–40. http://dx.doi.org/10.1016/j.proci.2004.08.251 , Jan-2005
Presentation	X. Qin and Y. Ju "Experimental and Numerical Study of Spectral Dependent Radiation Reabsorption on Flame Propagation" AIAA-2005-149, 43rd AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, Jan. 10-13, 2005 Jan-2005
Presentation	X. Qin, Z. Chen, and Y. Ju "Experimental and Numerical Study of Spectral Radiation in Propagating Spherical Flames" The 4th Joint Meeting of the U.S. Sections of the Combustion Institute Philadelphia, March 21-23, 2005. Mar-2005
Presentation	X. Qin, Z. Chen and Y. Ju "Effects of Radiation Reabsorption on Flame Propagation and Flammability Limits in CO2 Diluted Flames" Proceedings of 20th International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS), Montreal, Canada, Aug.1-5, 2005. Jan-2006